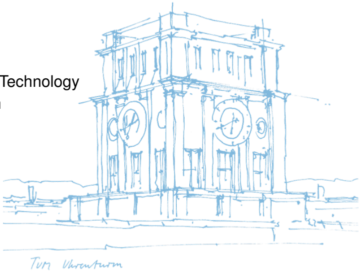
Chair of Connected Mobility
School of Computation, Information, and Technology
Technical University of Munich

An Overlay for Social Networks based on Directed Acyclic Graphs on the Edge

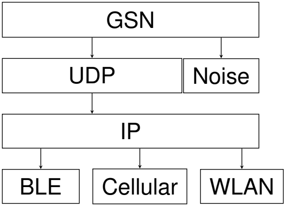
Dan Bachar, David Guzman

Sunday 17th May, 2026

Chair of Connected Mobility
School of Computation, Information, and Technology
Technical University of Munich

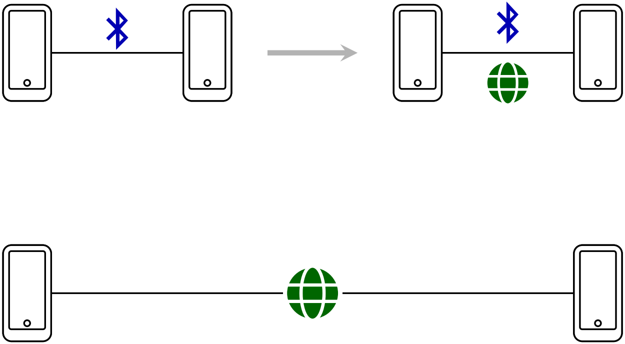


Architecture



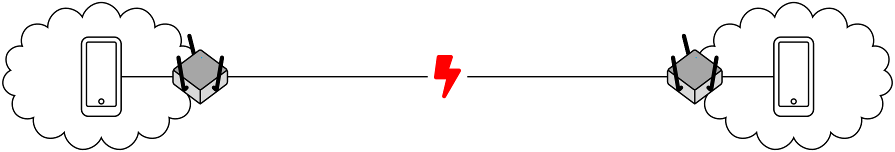
- Connect two peer devices according to priority and proximity pair
- Dual-stack network comms: BLE first, Cellular (5G/LTE) and WLAN second
- In close range, use BLE for signaling to establish UDP connection
- ICE-similar stack for address reflection, hole-punching using friends

Dual-stack comms

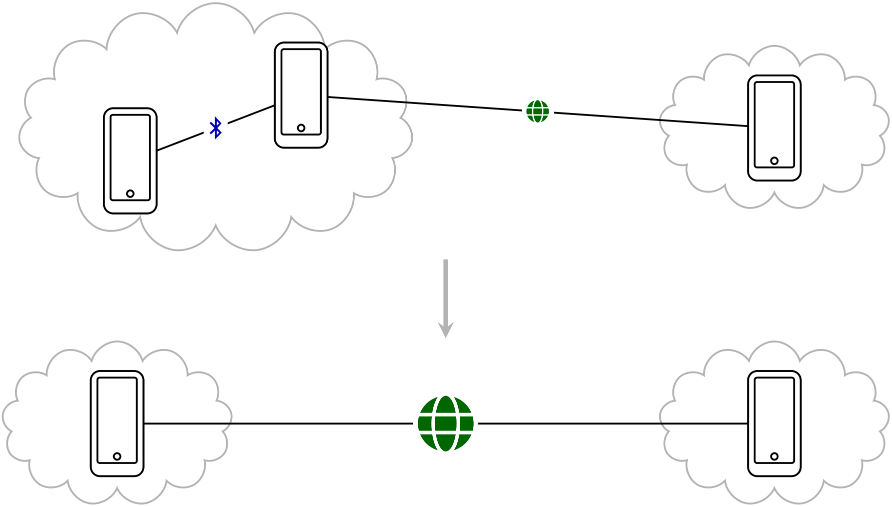


- BLE as signaling channel
- Advertise multiple addresses
- UDP for hole-punching
- UDX for long-range communication

NAT

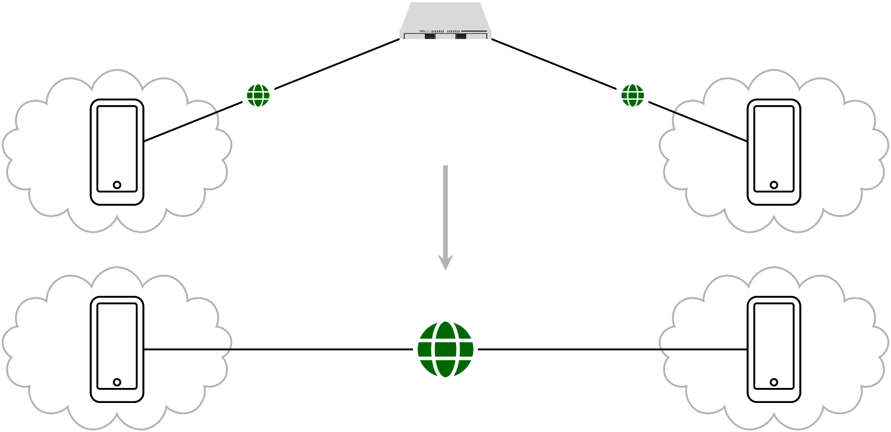


Well-connected friends



Chair of Connected Mobility
School of Computation, Information, and Technology
Technical University of Munich

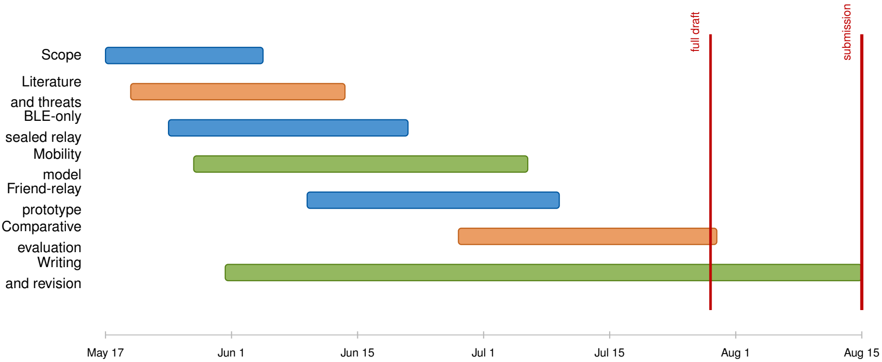
Rendezvous Server



Since mid-April: what we converged on

- Built the transport around cryptographic identity: Ed25519 public keys as peer identity and signed packets as the basic trust object.
- Stabilized the dual-stack direction: BLE discovery and nearby direct messaging, with UDP/UDX explored for longer-range connectivity.
- Worked through NAT traversal: public address discovery, hole-punching, well-connected friends, and rendezvous-style coordination.
- Added delivery machinery: ACK/read receipts, fragmentation for large payloads, Redux-backed peer/transport/message/friendship state.
- Added Noise sessions as hop-level channel security, then identified why relaying needs a separate end-to-end sealed message envelope.
- Reframed the thesis around a BLE-only path: remove UDP, analyze friend-based relaying, and compare it with direct BLE and open stranger relay.
- Connected the research plan to opportunistic networking, Sybil/misbehavior defenses, and Shapiro–Keidar work on grassroots/blocklace systems.

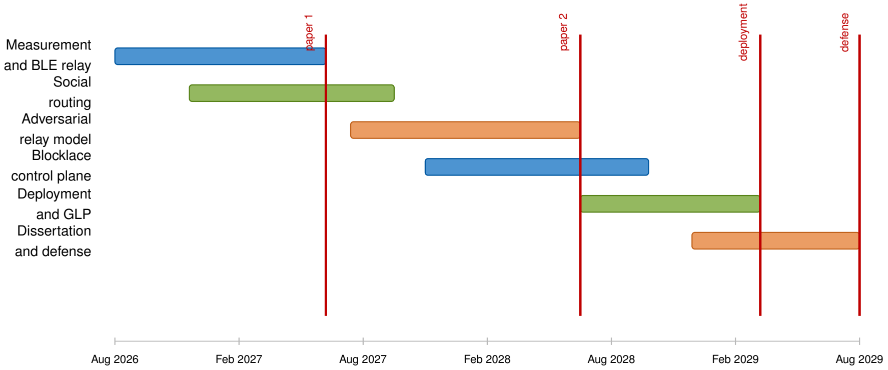
Master thesis gantt: submission on 15 Aug 2026



- Minimum contribution: secure relay design, mobility model, comparison, and critical security analysis.

Chair of Connected Mobility
School of Computation, Information, and Technology
Technical University of Munich

PhD runway: three-year research arc



- Thesis-to-PhD bridge: from "does BLE relay work?" to "can grassroots social trust make mobile relaying secure and abuse-resistant?"